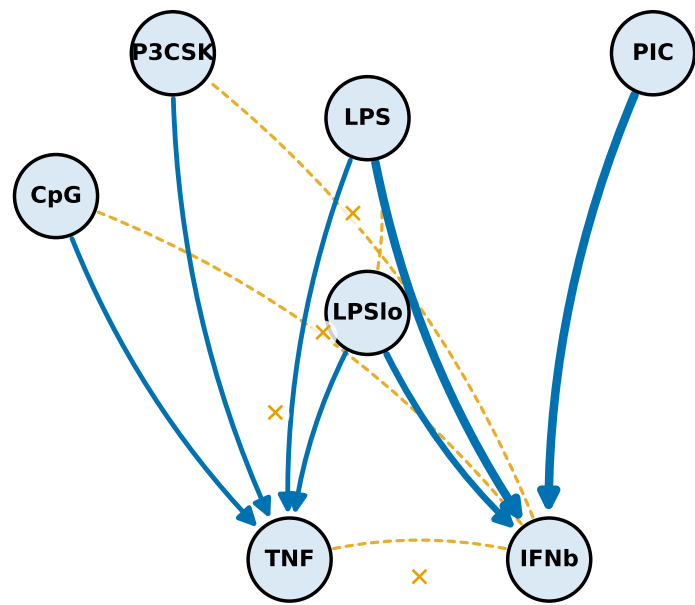
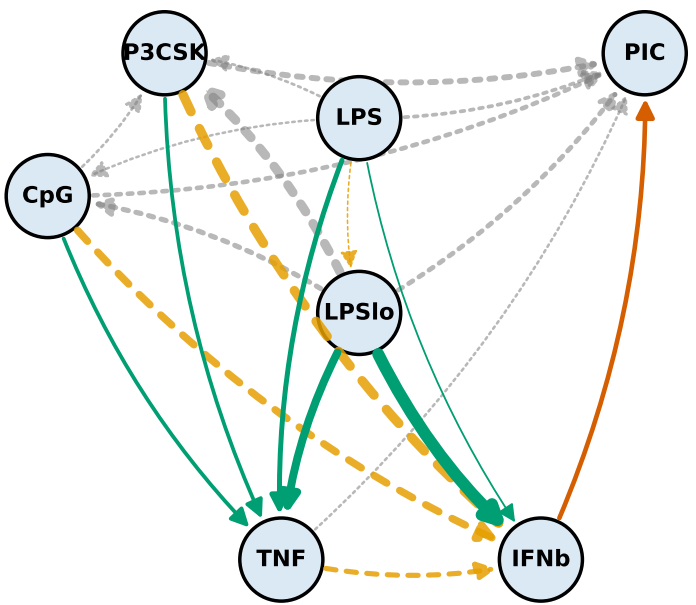


Sheu 2024 BMDM: real-data cascade-direction validation (analog of cascade_forge Fig. 1 + Fig. 3)

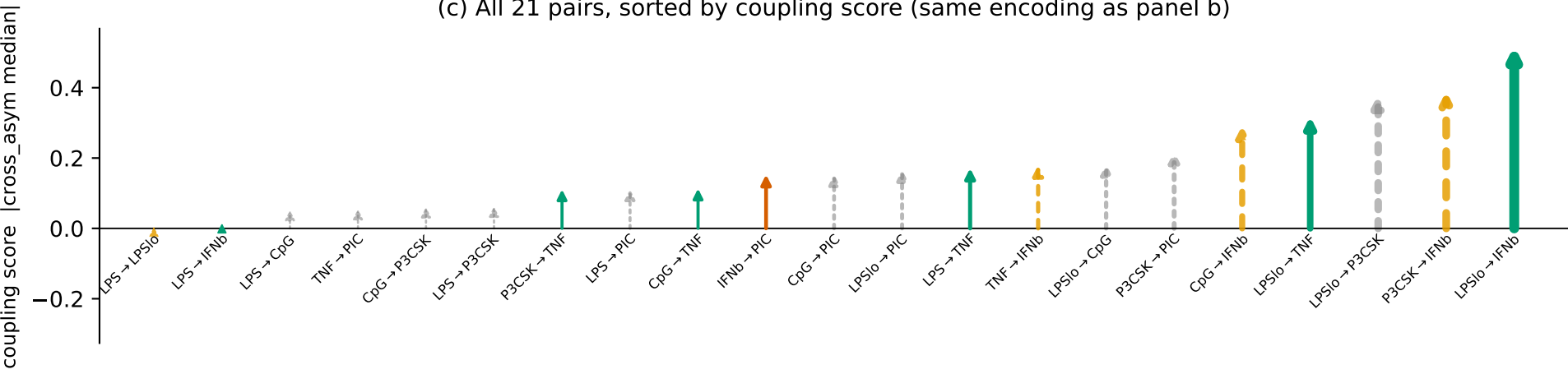
(a) Pre-registered ground truth
(Sheu TLR cascade wiring, §21/§26)



(b) What cross_asym found (Sheu, 5hr)
benchmark direction accuracy: 6/7 = 86%



(c) All 21 pairs, sorted by coupling score (same encoding as panel b)



— ground truth cascade edge
- - - ground truth: no cascade (negative control)
— found: correct direction (benchmark pair)
— found: wrong direction (benchmark pair)
· · · · found: no literature prior (uncalibrated)